EN 601.124
The Ethics of Artificial Intelligence and Automation:
AI and the future of work: Job displacement

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Announcements

1. Reading reflection due tomorrow (the last one!)
2. Fri 17th: project report due
3. Monday 20th: Project presentations
4. Wed 22nd: ePortfolio workshop (attendance not required if not relevant)
5. Fri 24th: ePortfolio submission due

Outline & goals

Today's goals:

1. Be able to answer questions such as:

- Does AI *really* make <**insert a profession**> more productive?
- Will AI reduce or increase the need for <**insert a profession**>?
- How does AI change what workers in <**insert a profession**> do?

2. Evidence-based reasoning about who benefits from AI productivity

3. Examine what ethical choices shape the future of work

Outline & goals

Outline:

1. **AI and productivity**
2. **AI and skills**
3. **Ethical lenses**
4. **In class activity**
5. **Need for innovation**

Three key ethical risks of AI on human work

- **Displacement** (job loss)
- **Degradation** (loss of autonomy, meaning, skill)
- **Extraction** (surplus captured by firms, not workers)

1) AI and productivity

Background: AI and Productivity

- Growth in productivity (output per unit input) is crucial for improving long-term living standards
- General-purpose technologies, defined as pervasive, improving over time, and spawning complementary innovations, are key to increasing productivity
- AI is general purpose technology
- The productivity effects of AI depend on its adoption and use in different sectors and tasks
- Given AI's wide applicability and the rate of its technical development, significant impacts on productivity are likely in the coming decade

- A Harvard Business School study found that low-skilled consultants gained more from AI systems than high-skilled consultants (43% faster versus 17% faster), even outperforming those without AI
- A University of Minnesota study observed that young attorneys could perform a variety of standard tasks on average 22% faster with AI systems.

Working Paper 24-013

Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality

Fabrizio Dell'Acqua
Edward McFowland III
Ethan Mollick
Hila Lifshitz-Assaf
Katherine C. Kellogg
Saran Rajendran
Lisa Krayer
François Candelon
Karim R. Lakhani

Lawyering in the Age of Artificial Intelligence

109 Minnesota Law Review (Forthcoming 2024)
Minnesota Legal Studies Research Paper No. 23-31

65 Pages • Posted: 9 Nov 2023 • Last revised: 15 May 2024

Jonathan H. Choi
Washington University School of Law

Amy Monahan
University of Minnesota Law School

Daniel Schwarcz
University of Minnesota Law School

Figure 1: Quality Distributions with and Without AI—Complaint Drafting

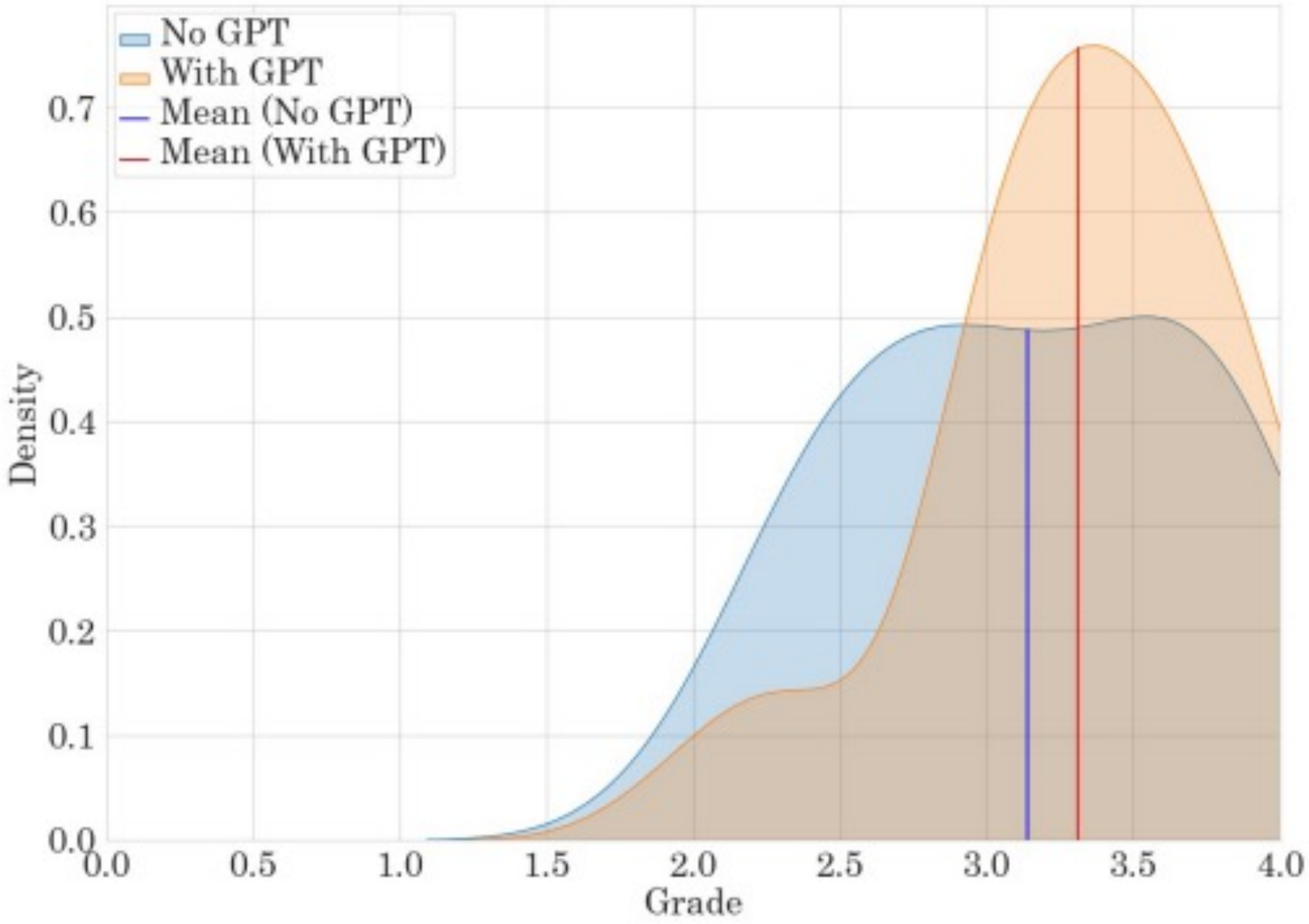


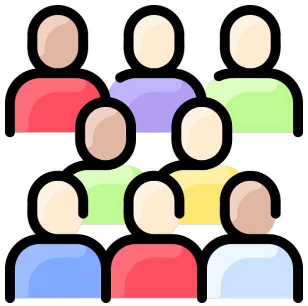
Table 2: Average Time Taken on Tasks with and Without GPT-4 (Minutes)

	No GPT-4 (Std. Dev.)	With GPT-4 (Std. Dev.)	Difference (95% CI)	% **Difference**	*p*-value
Complaint Drafting	160.69	122.00	-38.77	24.1%	0.0018
	(72.38)	(66.80)	(-64.00, -13.36)		
Contract Drafting	69.72	47.59	-22.40	32.1%	0.0000
	(32.00)	(31.09)	(-33.71, -10.91)		
EE Handbook	37.24	29.41	-7.84	21.1%	0.0000
	(9.55)	(13.42)	(-12.03, -3.74)		
Client Memo	244.41	215.69	-28.75	11.8%	0.0152
	(58.03)	(72.96)	(-52.59, -5.05)		

Detour (example from our recent research)

Participant panel

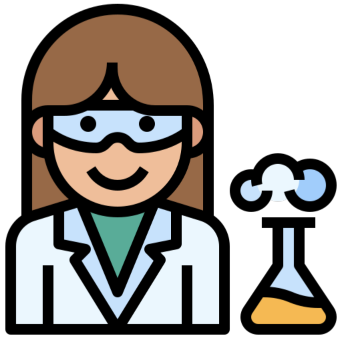
Provides feedback



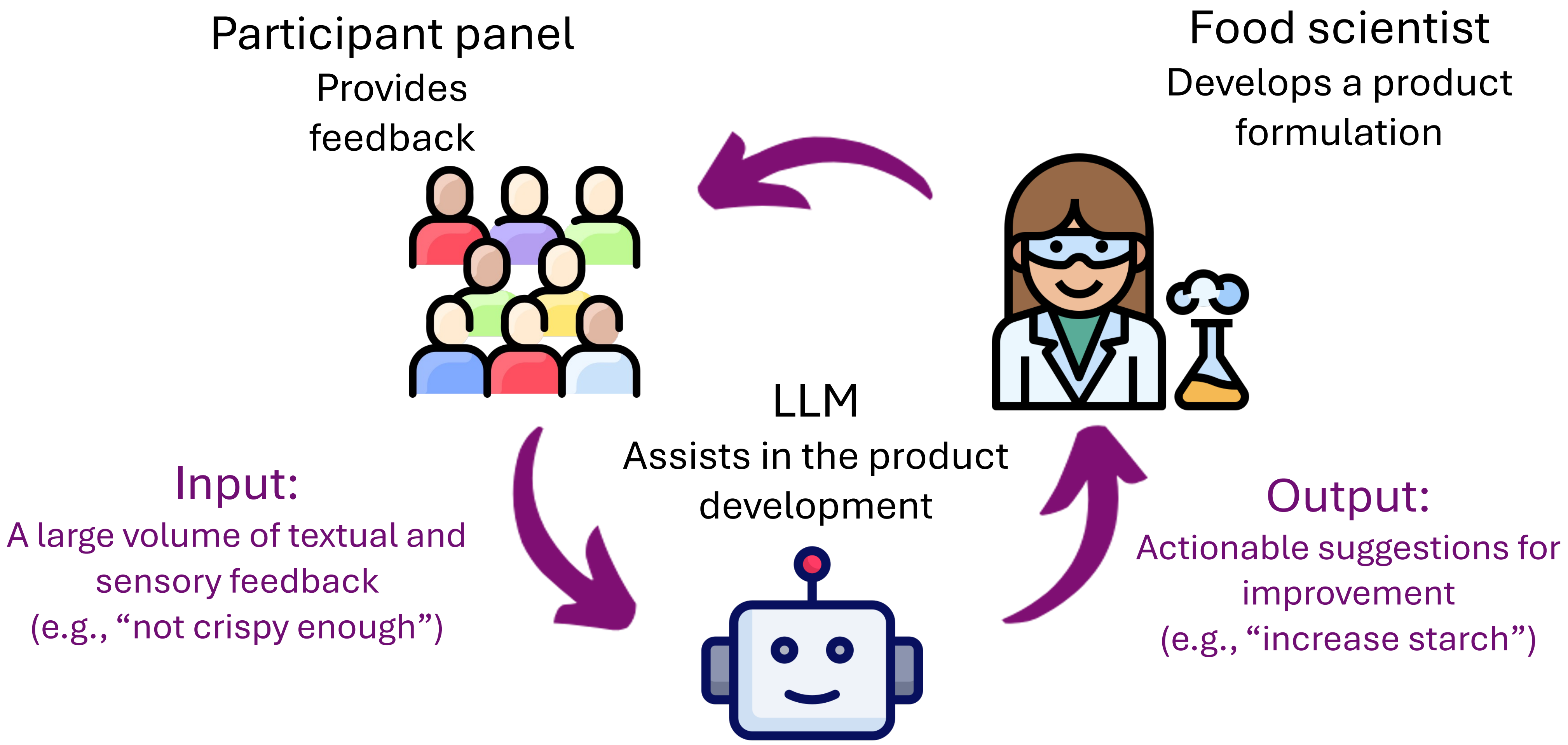
A large volume of textual and sensory feedback
(e.g., “not crispy enough”)

Food scientist

Develops a product formulation



- <u>Effectiveness</u>: Time consuming!
- <u>Creativity:</u> Boring!


Participant panel
Provides feedback
Food scientist
Develops a product formulation
LLM
Assists in the product development
Input:
A large volume of textual and sensory feedback
(e.g., “not crispy enough”)
Output:
Actionable suggestions for improvement
(e.g., “increase starch”)

Participant panel
Provides feedback

Food scientist
Develops a product formulation

A study with food scientists

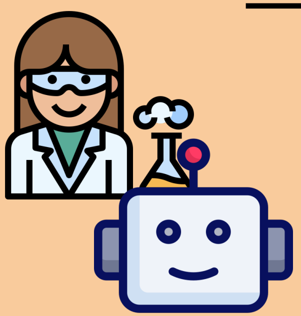
- 20 expert food scientists
- 30 products with detailed qual. and quant. feedback

Input:
A large volume of textual and sensory feedback (e.g., “not crispy enough”)

Output:
Actionable suggestions for improvement (e.g., “increase starch”)

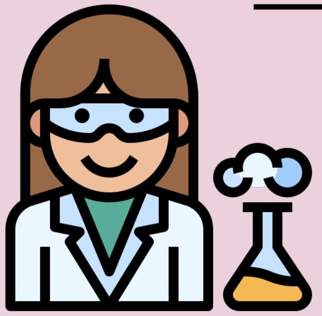
LLM-assistance in product design

Phase 1:



generate suggestions

Phase 2:



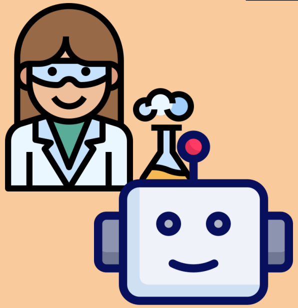
evaluate suggestions

What experiment would you consider next?

How accurate, specific, complementary, and helpful?

LLM-assistance in product design

Phase 1:



generate suggestions

What experiment would you consider next?

Input:

Heterogenous textual and sensory feedback

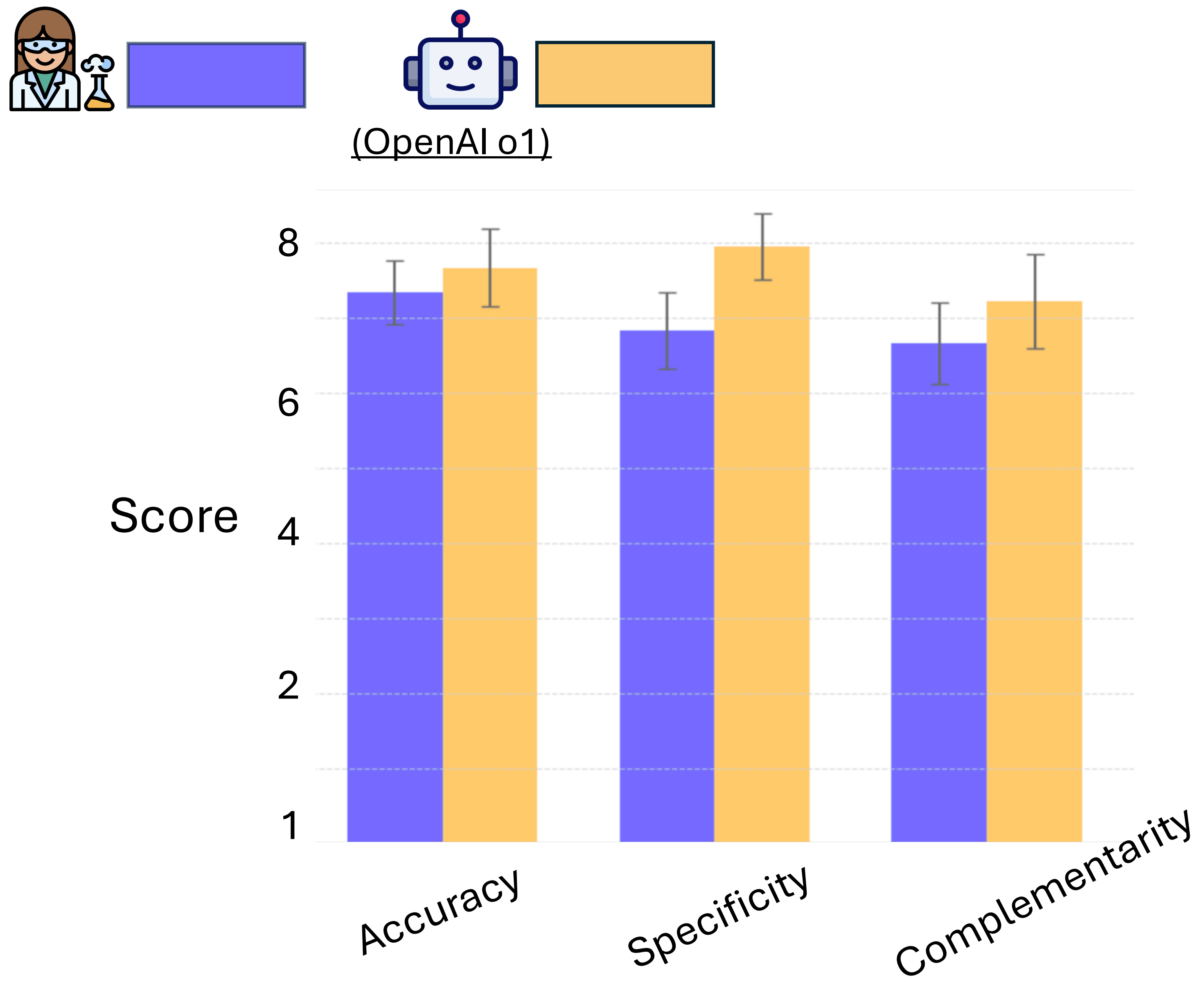
not crispy enough
not salty
after-taste
…

Meatiness	Sweetness	Greasiness
1	4	7
3	3	6
…	…	…

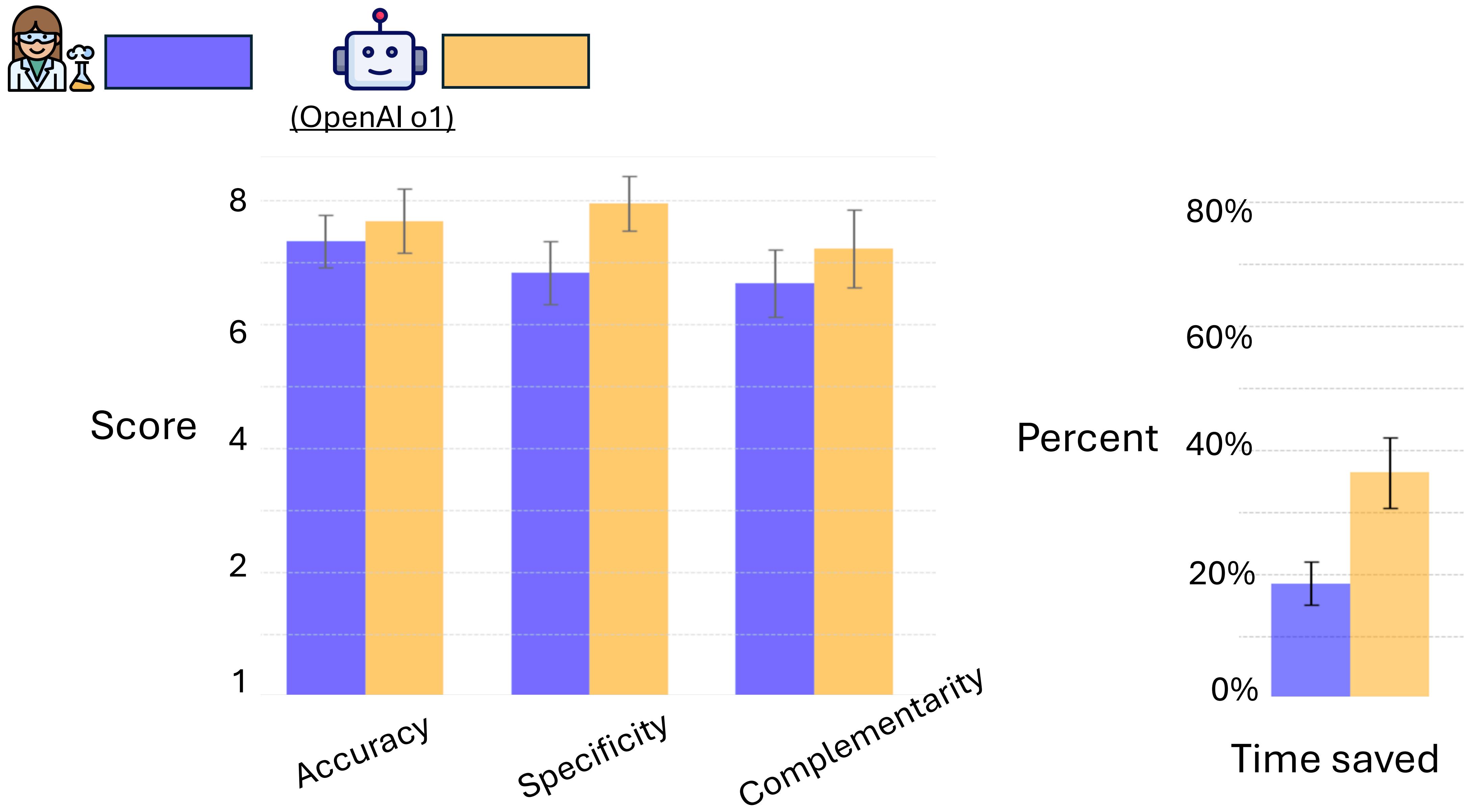
Output:

A suggestion

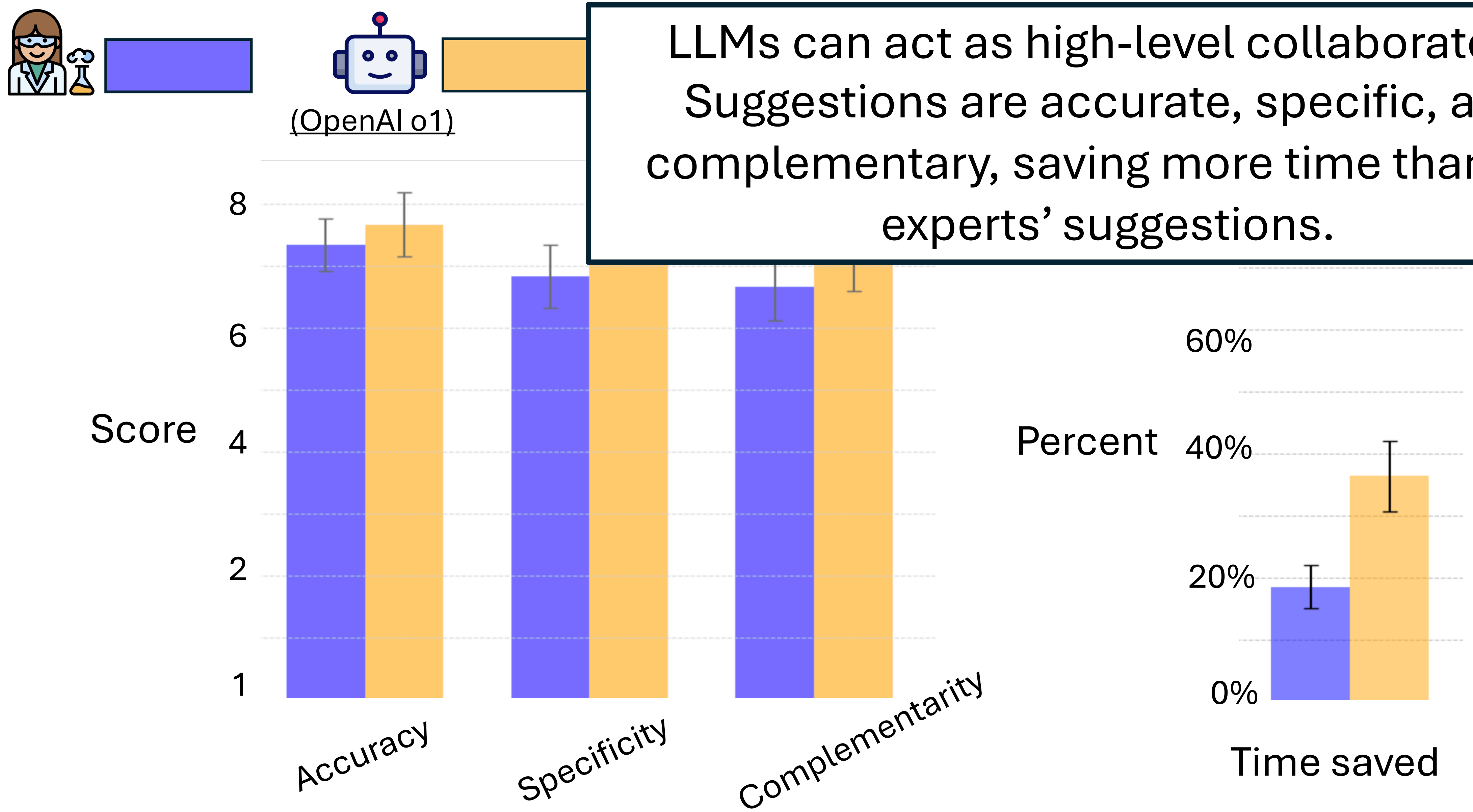
e.g., increase starch



(OpenAI o1)
Score
8
6
4
2
1
Accuracy
Specificity
Complementarity



(OpenAI o1)
Score
8
6
4
2
1
Accuracy
Specificity
Complementarity
Percent
80%
60%
40%
20%
0%
Time saved



LLMs can act as high-level collaborators: Suggestions are accurate, specific, and complementary, saving more time than the experts' suggestions.

What choices made this ethical?

- **Risk of displacement:** Tested an approach that can augment their skills, not replace
- **Disk of degradation**: Automate the “dreadful”, not “creative” tasks!
- **Risk of extraction:** Who benefits from the surplus? Engaged with affected professionals (start-ups)
- Robust evaluation & evidence-based approach
  - Separate sets of experts
  - Randomized trial

Potential Effects of increased productivity (1)

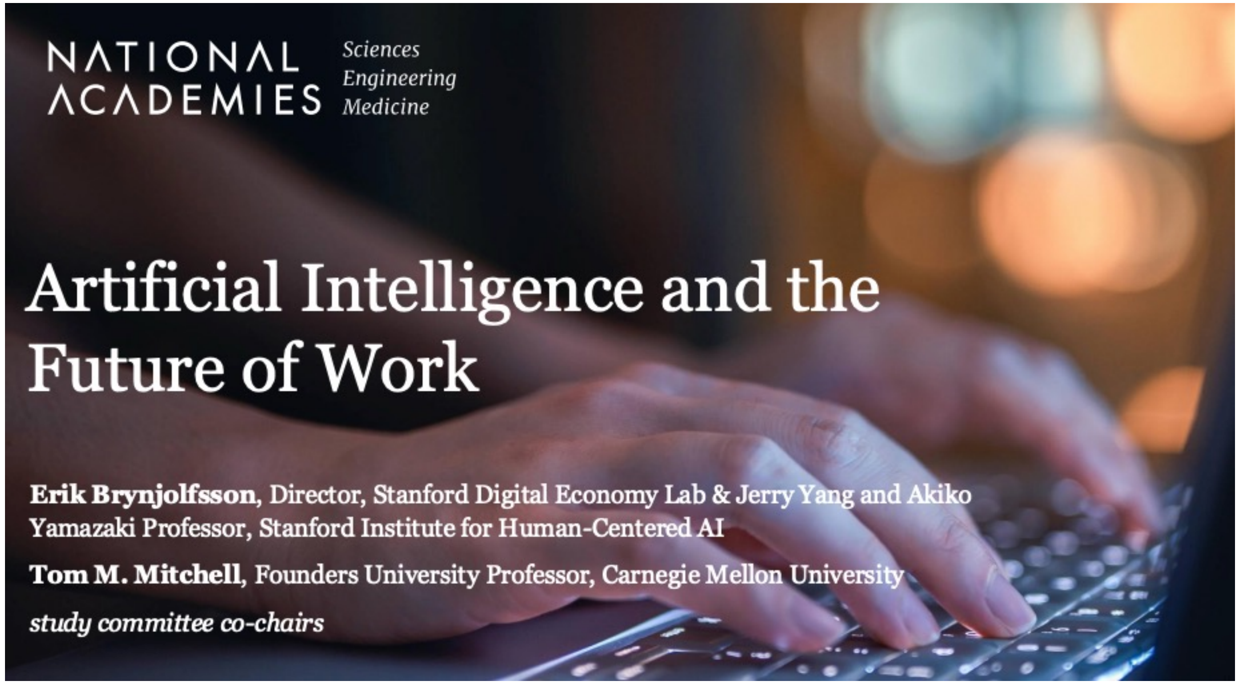
- Generative AI has already increased productivity in specific applications including contact centers, software development, and writing. Many other tasks not suitable for at least current forms of AI
- It is difficult to predict how AI could impact productivity, but some estimates suggest the rate of growth of the U.S. productivity could more than double from about the recent rate of 1.4 percent to 3 percent or more
- AI’s potential to accelerate scientific discovery and innovation could further compound productivity gains
- There is uncertainty about the timing and magnitude of any productivity gains

Potential Effects of increased productivity (2)

- Without institutional and policy changes, the benefits of productivity might not be widely shared. This could lead to job losses and wage disparities, increased inequality, and adverse effects on job quality and worker satisfaction
- Productivity is not the only measure of human well-being. It is important to also consider and measures how AI might affect other aspects of human well-being

Research on the Potential Impacts of AI on Productivity

- As was the case with earlier general-purpose technologies, achieving the full benefits of AI will likely require complementary investments in new skills and new organizational processes and structures



Artificial Intelligence and the Future of Work. Erik Brynjolfsson, Tom M. Mitchel

2) AI and skills

Demand for Expertise

Expertise: A specific body of knowledge or competency required to accomplish a particular objective.
Examples: baking bread, taking vital signs, coding an app

The key question is how AI will alter the demand for different types of worker expertise.

New technology could:

1. **Erode** the value of existing types of expertise (e.g., tax preparation software erodes the value of tax expertise) or

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The creation of demand for new expertise is a critical force that counterbalances the tendency of new technology to erode the value of old expertise.

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Historical analogs:

Number of farmers
Agricultural industrialization post WWII

Elastic demand:
Productivity gains means more jobs!

Inelastic demand:
Productivity gains means jobs will be lost!

If product demand is sufficiently elastic, productivity-enhancing technology **increases** industry employment.

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Number of commercial pilots
Jet engines and autopilot systems ~1960s

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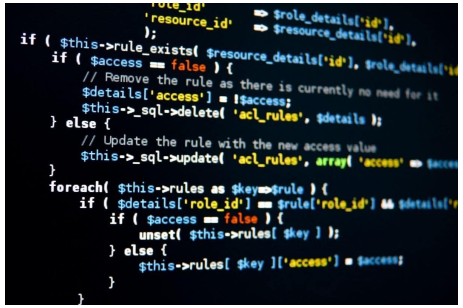

Number of programmers
The digital revolution (microprocessors and PCs)

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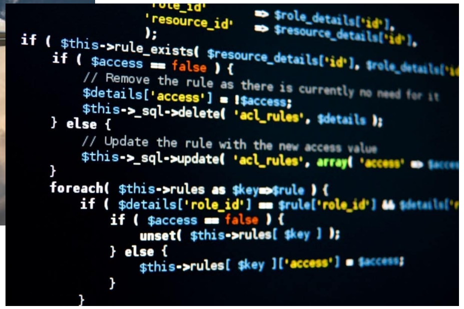
Historical analogs:

Air travel



Programming

Lawyers?
Educators?
Therapists?
Doctors?
Scientists?
…
Large language models

Agriculture

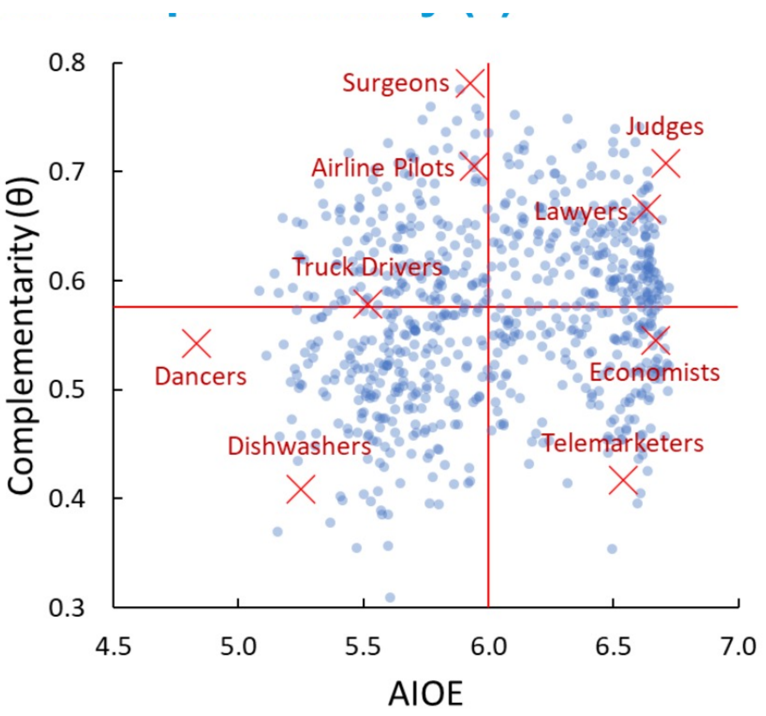
Elastic demand:
Productivity gains means more jobs!

Inelastic demand:
Productivity gains means jobs will be lost!

If product demand is sufficiently elastic, productivity-enhancing technology **increases** industry employment.
***Caveat: History does not repeat — institutions determine outcomes.**

Alternative lens: Asking people!

How much AI *augments* rather than replaces the worker. Higher = AI makes the human more productive.



How much a job's tasks overlap with what AI can do. Higher = more exposed to AI substitution.

Mauro Cazzaniga, Florence Jaumotte, Longji Li, Giovanni Melina, Augustus J. Panton, Carlo Pizzinelli, Emma Rockall and Marina M. Tavares (2024). Gen-AI: Artificial Intelligence and the Future of Work." *IMF* (2024).

Jobs in U.S. that are likely to have high, medium or low exposure to AI

High exposure
- Budget analysts
- Data entry keyers
- Tax preparers
- Technical writers
- Web developers

Medium exposure
- Chief executives
- Veterinarians
- Interior designers
- Fundraisers
- Sales managers

Low exposure
- Barbers
- Child care workers
- Dishwashers
- Firefighters
- Pipelayers

Note: Occupations are grouped by the relative importance of work activities with low, medium or high exposure to AI.
Source: Pew Research Center analysis of O*NET (Version 27.3).
"Which U.S. Workers Are More Exposed to AI on Their Jobs?"

PEW RESEARCH CENTER

Scenarios for the Impact of AI on Demand for Expertise

Scenario	Likelihood
AI accelerates occupational polarization, automating more nonroutine tasks and increasing the demand for elite expertise while displacing middle-skill workers	Plausible based on earlier technologies, but new technologies appear to be different
AI advances to outcompete humans across nearly all domains, greatly reducing the value of human labor and creating significant income distribution challenges	Unlikely in the near future owing to the limitations of AI, demographic trends, and potential for emergence of new forms of work

Artificial Intelligence and the Future of Work. Erik Brynjolfsson, Tom M. Mitchel

Technological Change and Demand for Expertise (1)

In prior major technological transitions, important categories of previously valuable human expertise were stranded by new machines and novel forms of work organization.

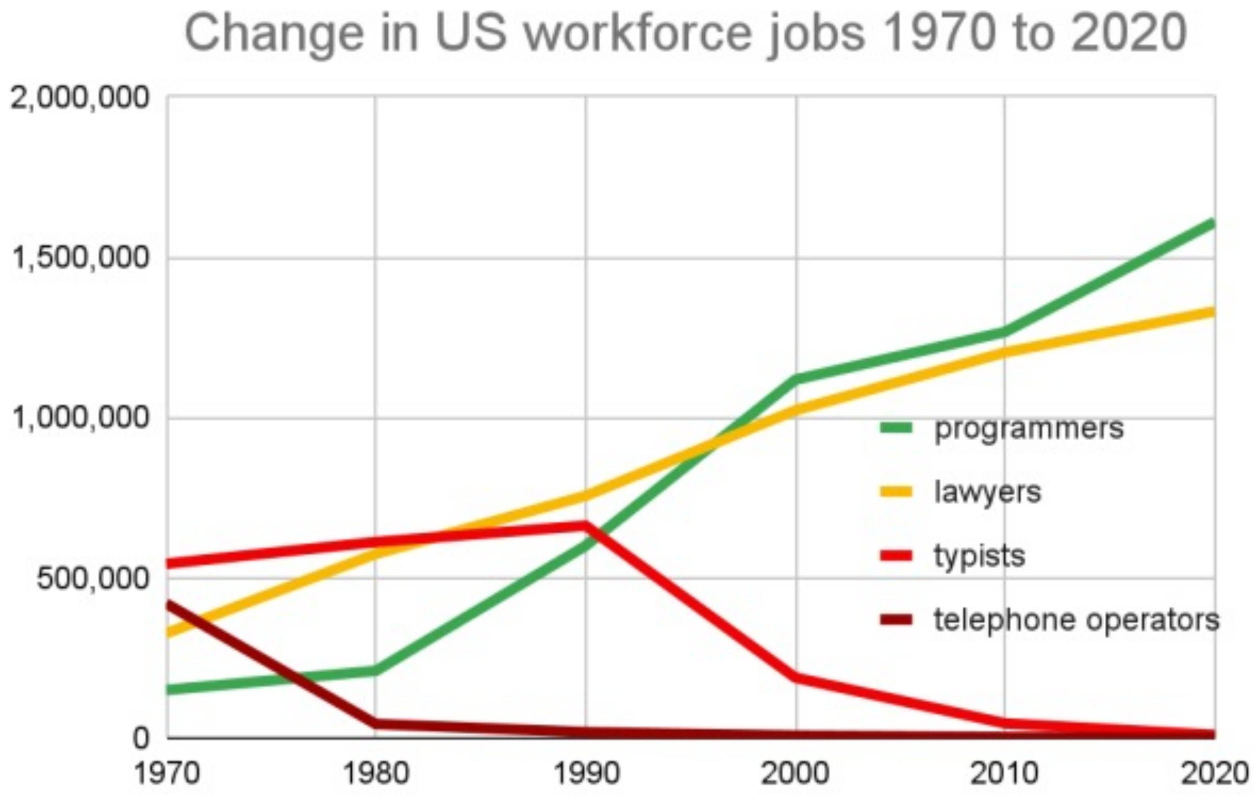
1. The value of artisanal skills was eroded by the rise of the factory system.

2. The value of skills in repetitive clerical and production tasks was eroded by computerization.

Technological Change and Demand for Expertise (2)

- Over the longer run, new forms of expertise gained value and catalyzed job creation, although often benefiting a different set of workers from those who were displaced.
- Indeed, recent research estimates that more than 60 percent of current employment is found in occupational specialties that were not present in 1940.
- Substantial uncertainty about what types of new work will follow from the widespread use of AI, what skills it will require, what it will pay, how much of it there will be, and who will do it.

Technological Change and Demand for Expertise (2)



The number of telephone operators, typists, lawyers, and programmers over 50 years based on U.S. Census data for each decade since 1970. The number of typists fell ~50 fold and telephone operator jobs shrank ~300 fold, while the number of lawyers grew ~fourfold and programmers shot up ~11 fold. (The U.S. population grew by 60% over that period.) Lawyers now use their computers by themselves, type their own emails and texts, and use their own smartphones. In the law office, a lawyer—plus programmer-developed technology—replaced the typists and telephone operators. In 1970, the ranking by the number of jobs were typists, telephone operators, lawyers, and programmers, which was nearly reversed by 2020. Technology advances were behind the shift in these jobs.

Questions to Ask about the Impact of AI Advances on Jobs (1)

- What expertise will be substituted with or made obsolete by AI?
  - AI may soon surpass human capabilities in a variety of tasks requiring elite expertise, such as producing documents and illustrations, or managing complex systems.
  - Continued progress in robotics will substitute for widely available physical expertise, such as package delivery or high-volume food preparation.

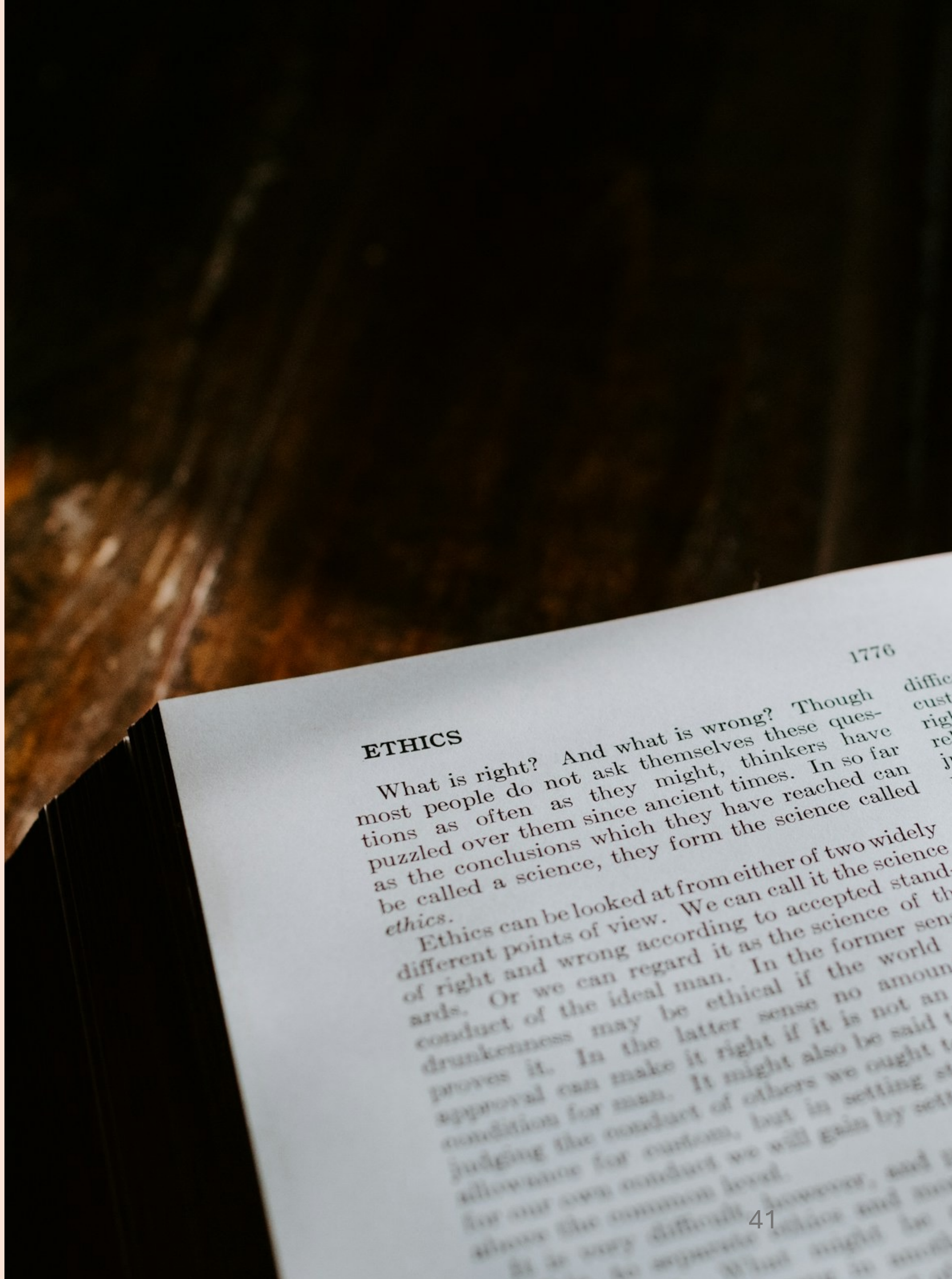
Questions to Ask about the Impact of AI Advances on Jobs (2)

- What expertise will be augmented, or newly demanded, as a result of AI adoption?
  - AI has the potential to complement human expertise in some jobs, for example:
    - Doctors making diagnoses
    - Architects developing and evaluating a wider range of building designs

Questions to Ask about the Impact of AI Advances on Jobs (3)

- How feasible will it be for workers to acquire newly valuable expertise?
  - Impacts depend on what types of expertise are required, how much time and effort to acquire, and whether retraining and certification opportunities are available
  - Will government subsidize the cost?

3) Five ethical lenses to reason about the ethics of AI on workforce



(1) AI & Task recomposition

- **Task recomposition**: AI doesn't just automate tasks; it *rearranges* who does what, when, and with what autonomy.
- Leads to:
  - deskilling of some tasks
  - intensification of remaining human labor
  - loss of end-to-end ownership
- **Why it matters ethically**
  - Workers may keep jobs but lose *meaning*, autonomy, or career ladders.
  - Raises questions of **job quality**, not just job quantity.

(1) AI-driven task recomposition

- Software engineers spending less time designing systems and more time:
- reviewing AI-generated code
- debugging subtle failures
- handling edge cases and blame

Acemoglu & Restrepo (2020): *Tasks, Automation, and the Rise in Wage Inequality*

(2) AI-driven surveillance

- AI not just as a productivity tool, but as a **managerial technology**
- Includes:
  - keystroke logging
  - productivity scoring
  - automated performance evaluation
  - algorithmic scheduling / firing
- **Why it matters ethically**
- Power asymmetry: workers are evaluated by opaque systems they don’t control
- AI may be used to monitor worker activity and productivity: a potential tool to better link individual worker performance and compensation but also an enabler of surveillance that workers may find objectionable.
- Chilling effects on creativity, dissent, and experimentation

(3) The AI-driven “illusion of competence”

- AI increases speed → users *believe* they are more competent; but they over-rely on AI
- But may:
  - reduce learning
  - mask errors
  - erode deep expertise over time
- **Why it matters ethically**
- Safety-critical domains (medicine, law, infrastructure)
- Long-term skill atrophy vs short-term productivity
- **Key studies**
  - Noy & Zhang (2023): ChatGPT increases productivity but homogenizes outputs
  - Studies showing **novices benefit most**, but experts risk over-trust

(4) Who captures the surplus? AI-exacerbated inequalities

How will the organization of work and employer-power dynamics evolve?

- Outcome for worker and societal welfare necessarily depends on the legal, regulatory, and bargaining regimes in place that will shape design, adoption, and use of AI, and the distribution of economic surplus among owners, managers, and line workers.

(4) Who captures the surplus? AI-exacerbated inequalities

AI productivity gains ≠ worker wage gains

- Surplus can go to:
  - firm owners
  - platform providers
  - model developers
  - cloud infrastructure companies
- **Ethical question to pose explicitly**
  - If AI makes workers twice as productive, who gets the value?

(5) AI-caused and labor offshoring

- AI changes *where* work happens, not just whether it happens
- Some work:
  - reshored (automation)
  - offshored (AI-mediated gig work)
  - invisibile (data labeling, content moderation)
- **Concrete examples**
  - Global data annotation labor
  - Content moderation in the Global South
  - Prompt-engineering as invisible labor
- **Ethical questions**
  - Who bears the cognitive and psychological costs?
  - What labor protections apply across borders?

In-class activity

In-class role-play activity: **Should I Become a Software Engineer in the Age of AI?**

Student 1: The worried high schooler

You are a high school student who wants to become a software engineer. You are worried that AI tools (like ChatGPT, GitHub Copilot, etc.) will make it pointless to study computer science.

Your goal: Clearly express your concerns and ask tough questions about the future of software engineering.

Students 2 and 3: Encouraging friends

You believe software engineering will *remain* a strong and evolving career. You are trying to convince your friend not to give up on becoming a software engineer.

Your goal: Respond thoughtfully to concerns and provide realistic, balanced reasons to stay in the field.

Instructions

1. Spend **3 minutes** preparing your arguments.
2. Act out the conversation for **10 minutes**.
3. Stay in character and respond naturally to each other.

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Debrief

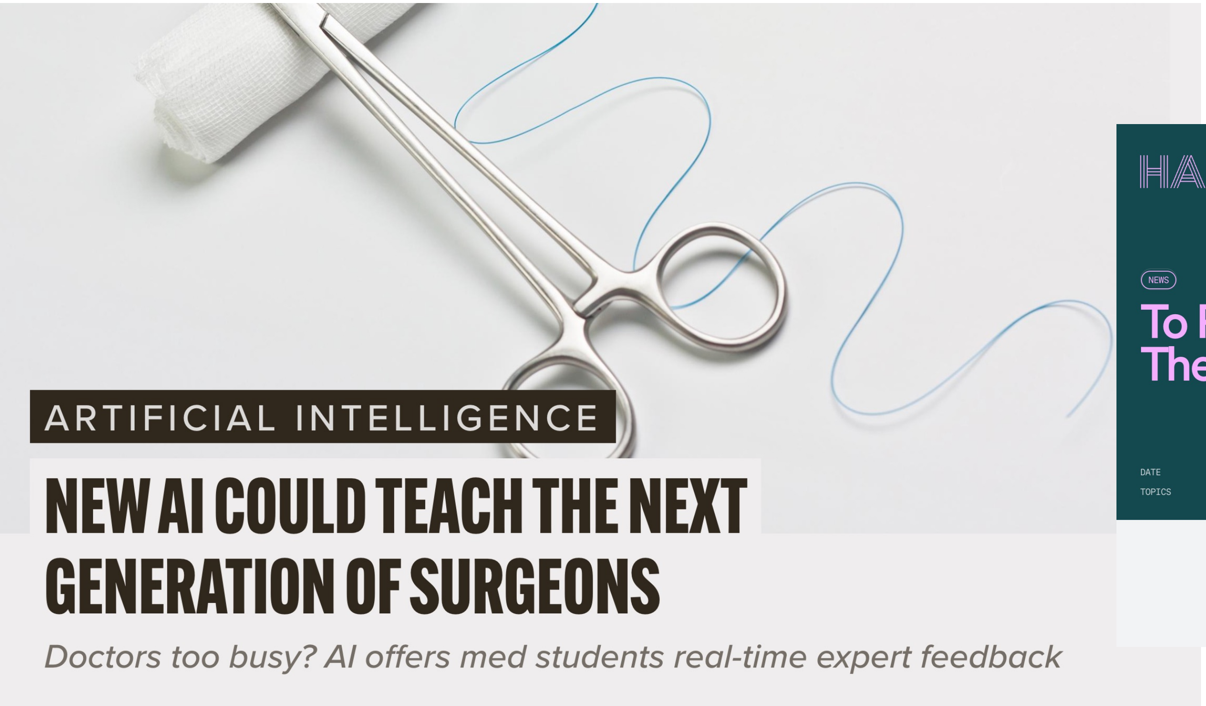
What fears about AI felt most convincing? Who “won” the argument?

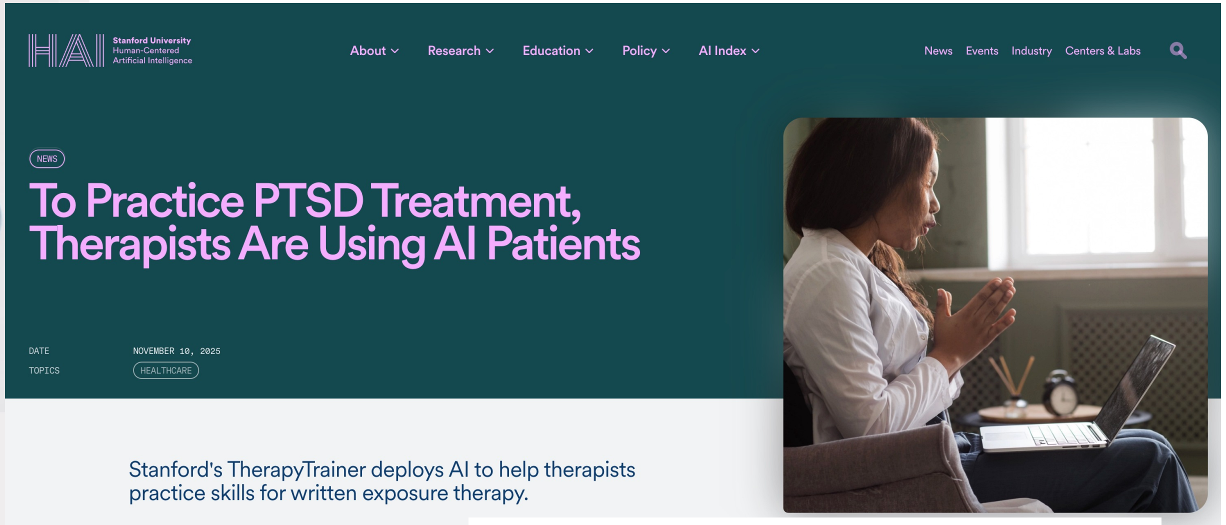
- What arguments were most persuasive?
- How might the role of a software engineer change rather than disappear?
- What skills should future engineers focus on learning?

5) Looking ahead: the need for innovation

Opportunities to Balance Worker Augmentation, New Work Creation, and Labor-Displacing Automation

1. **Advocate for:**
   a) **Augmenting instead of replacing**: AI that complements humans and improves quality.
   b) **Participatory values**: Best practices for fostering inclusive AI adoption within organizations and strengthening worker voices in business decisions.
   c) **Evidence-based approaches**: How the relative cost of capital and labor affects business decisions about AI adoption for automation vs. for augmentation.
   d) **Regulation:** New norms or legal protections to protect workers rights to control and be compensated for the use of their likenesses, their other personal attributes, and their creative works.
2. Increase AI expertise within the federal government to support effective investment, oversight, and regulation across all mission areas.
3. Evaluate the quality of human-complementary technology before adopting it for publicly funded programs in such areas as education and health care.





HARVARD Kennedy School
BELFER CENTER
for Science and International Affairs

EVENT SUMMARY

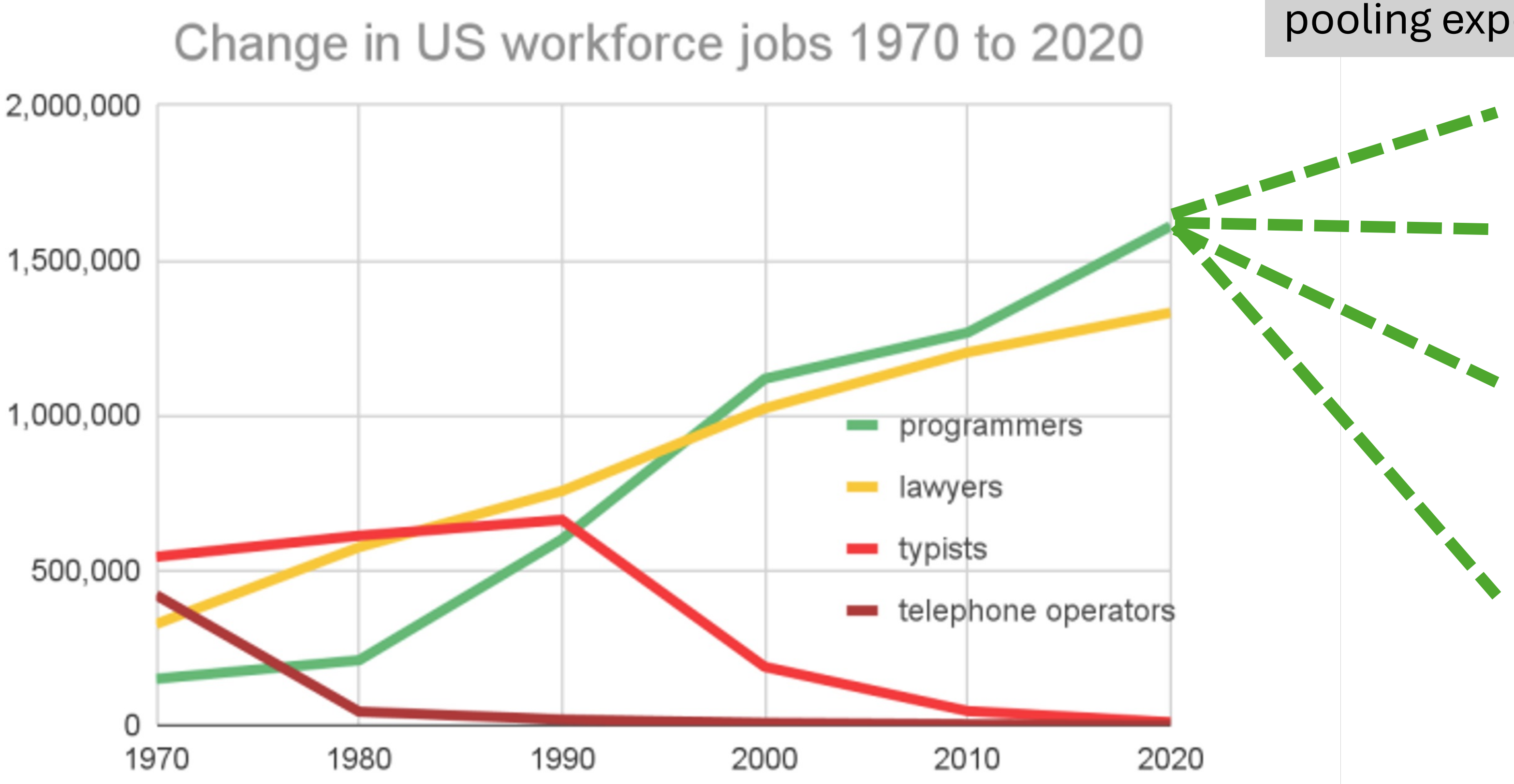
AI and the Future of Conflict Resolution: How Can Artificial Intelligence Improve Peace Negotiations?

AI Milestone: Rapid Upskilling. Many current workforce development programs have a modest impact on wages. A milestone: an AI system that could reduce the time for workers who are unemployed or in low-income jobs to gain an in-demand skill that is a ticket to the middle class. In the U.S., for example, a goal might be to empower a worker making ≤$30,000 annually to gain a skill that increases their income by ≥$15,000, and to do so in three to six months.

AI Milestone: Job Forecaster. To help workers design their career roadmaps, we need to observe and communicate workforce changes. A Job Forecaster would aid workers displaced by AI systems to retrain for well-compensated jobs with growth potential. A real-time Job Forecaster could guide them to an educational path to retrain for promising jobs with future growth.

What happens next?

Learn from online signals, job ads, social media, news, pooling experts, etc.?



Research on AI and Education

- AI will have significant implications for education at all levels, from primary education, through college, through continuing education of the workforce:
  - Driving the demand for education in response to shifting job requirements.
  - Driving the supply of education as AI provides opportunities to deliver education in new ways.
  - Shifting what we should teach to the next generation to prepare them to take full advantage of future AI tools and advances.

Conclusions

AI's impact on the workforce is not inevitable: institutions and design choices matter!

- Engineers (design choices)
- Firms (deployment incentives)
- Policymakers (institutions)
- Workers (adoption, resistance, retraining)

Conclusions

- **What Should *You* Take Away?**
- AI won’t just replace jobs, it reshapes them
- Skills ≠ tools; judgment and accountability matter
- Ethics isn’t about stopping AI, but shaping incentives
- Your career choices influence how AI is used

Readings & references

- Acemoglu & Johnson: *Power, Progress, and the Future of Work*
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